ChatGPT

Hospital Healthcare Management System – Power BI Project

Capgemini Healthcare Analytics Initiative
Your Name
Duration: Sept 2021 – June 2023

Table of Contents

Executive Summary

Figure: Sample placeholder image of a healthcare analytics scenario. The Hospital Healthcare Management System project at Capgemini involved creating a suite of **six** Power BI dashboards to transform raw hospital data into actionable insights. Over the 20-month engagement (Jan 2022 – Jun 2023), I delivered interactive reports for **Overview**, **Patient**, **Doctor/Commission**, **Operations**, **Finance**, and **Staff** domains. Each dashboard consolidated multiple data sources (EHR, billing, HR, etc.) to present key metrics such as patient volumes, bed occupancy, revenue, and staffing in real-time. By unifying diverse data into a single view, the solution enabled quicker decision-making and better resource management [1] [2]. For example, executives could instantly see hospital-wide KPIs and drill into departments, aligning with best practices where “top-level” alerts are shown first [3]. The project improved reporting efficiency (eliminating manual reports) and provided a clear, unified dashboard suite for clinical and administrative stakeholders.

Project Initiation & Planning

- **Scope Definition:** We began by conducting *stakeholder interviews* with hospital administrators, doctors, and finance teams to understand key challenges (e.g., high ER wait times, revenue leakage). As advised, we “narrowed focus to real problems” such as patient flow and billing bottlenecks [4]. This ensured clear objectives (e.g., reduce ER wait time) that drove KPI selection and data needs.
- **Methodology:** Using Agile sprints, we broke the project into iterations. Initial planning sprints (Sept–Dec 2021) defined requirements and data sources, while development sprints (Jan 2022 onward) delivered dashboard prototypes and refinements. Regular demos were scheduled with stakeholders for feedback loops.
- **Team & Tools:** The project team included myself (Power BI developer), a data engineer, and domain experts. We standardized on **Power BI Desktop** for report building and **Azure SQL** (on-prem gateway) for data storage. Power BI Service (Power BI Pro licenses) was used for deployment.
- **Planning Activities:** By Dec 2021 we had: (1) an inventory of data sources, (2) drafted star-schema data model, and (3) wireframes of each dashboard. We also set up development environments (Power BI Desktop on Windows, Git for version control) and a timeline (charting tasks each sprint). This planning phase ensured alignment and a solid foundation before development started in Jan 2022.

Data Foundation

Figure: Placeholder image for data analysis (Doctor examining reports). The data foundation involved integrating disparate hospital data into a **clean, unified model**. We identified key **source systems**: an EMR/EHR system (Epic/Cerner), SQL Server databases (labs, inventory), Excel/CSV exports (department logs, surveys), and third-party APIs (insurance claims) [5]. We used Power Query (M) to extract and transform each source. For example, patient data was loaded from an Excel sheet and cleaned with code like:

```
let
    Source = Excel.Workbook(File.Contents("HospitalData.xlsx")),
    Patients_Sheet = Source{[Item="Patients",Kind="Sheet"]}[Data],
    #"Promoted Headers" = Table.PromoteHeaders(Patients_Sheet,
[PromoteAllScalars=true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers",
        {{"PatientID", Int64.Type}, {"AdmissionDate", type date}, {"Age",
Int64.Type},
         {"Department", type text}, {"LengthOfStay", Int64.Type}})
in
    #"Changed Type"
```

Key steps: type conversions, renaming columns, merging queries, and handling nulls. After cleansing, we built a **star schema** model: fact tables (e.g. Admissions, Transactions) linked to dimensions (Patients, Doctors, Date, Departments). We ensured robust **relationships** (one-to-many) and included a full Date dimension for time intelligence. (As noted, “your Power BI dashboard will only be as good as your underlying relationships and DAX logic” [6].) Measures (DAX) were created for KPIs (see next sections).

Dashboard Development

We delivered **six** dashboards, each tailored to a domain. Consistent design themes (colors #003366, #0066CC, #FF6600) and layout principles (clear KPIs, logical flow) were applied. Key visuals included cards, line/bar charts, matrices, maps, and slicers. Below is a summary of each dashboard’s purpose and features:

Overview Dashboard

Figure: Placeholder for the Overview Dashboard with hospital KPIs. This “command center” dashboard provides a **high-level snapshot** of hospital performance (overall admissions, current in-patients, total revenue, etc.). It uses large KPI cards and trend charts to highlight urgent metrics. For example, a card shows *Current Inpatients* and a map visualizes occupancy by location. Line charts track monthly admissions and financial revenue. Slicers allow filtering by date range or department. This layout follows best practices: top-level alerts (e.g. if avg ER wait > target) at the top, followed by trends and filters [3]. The Overview enables executives to spot anomalies at a glance – consistent with the principle that dashboards must enable decision-makers to understand status in seconds.

Patient Dashboard

This dashboard drills into **patient-centric analytics**. It shows patient counts by department, demographics, and stay durations. Key visuals include:
- *Admissions by Month* (line chart)
- *Patient Demographics* (age/gender distribution pie charts)
- *Average Length of Stay* (gauge or KPI card)
- *Readmission Rates* (bar chart by department).

We created measures like:

```
AvgStay = AVERAGE('Patients'[LengthOfStay])
```

to compute average hospitalization time. Users can filter by date or condition. A drill-through enables clicking a row (e.g. Dept. Name) to see underlying patient records. This empowers clinical managers to identify trends (e.g. rising readmissions) and patient flows.

Doctor & Commission Dashboard

Focused on **provider performance and incentives**, this dashboard displays doctor-level KPIs. Visuals include:
- *Total Patients per Doctor* (bar chart)
- *Average Diagnosis Time* (card or box plot)
- *Commission Earned* (bubble chart comparing patient load vs commission).

We implemented DAX to calculate commission. For example:

```
TotalCommission = SUMX(
    Appointments,
    Appointments[Fee] * RELATED(Doctors[CommissionRate])
)
```

This multiplies each appointment fee by the doctor's commission rate. Another measure, **PatientsPerDoctor**, counts patient visits per physician. Filters allow viewing by specialty or month. This helps hospital admins reward high-performing doctors and manage referral patterns.

Hospital Operations Dashboard

This page tracks **operational metrics** that affect day-to-day management. Key indicators include: ER Wait Time, Bed Occupancy Rate, and Equipment Utilization. Visuals:
- *Average ER Wait Time* over hours (line chart)
- *Bed Occupancy* (gauge or donut chart showing occupied vs total beds)
- *Equipment Status* (matrix of devices and maintenance alerts).

For example, we calculated ER wait time with DAX:

```
AvgERWait = AVERAGEX(Visits,
    DATEDIFF(Visits[CheckInTime], Visits[TriageTime], MINUTE)
)
```

We applied color thresholds (red/yellow/green) to highlight critical values. This dashboard allows operations managers to quickly see if staffing or resources need adjustment.

Finance Dashboard

Figure: Placeholder for the Finance Dashboard (financial metrics and analytics). The Finance dashboard presents **financial KPIs**: monthly revenue, expenses, and profitability. Key visuals include a multi-row card for *YTD Revenue*, *YTD Expenses*, and *Profit Margin*, and a waterfall chart showing budget vs actuals. We used time-intelligence measures, for example:

```
YTD_Revenue = CALCULATE(
    SUM(Finance[Revenue]),
    DATESYTD('Date'[Date])
)
```

This aggregates revenue year-to-date. Other visuals: *Claim Denials by Month* (bar chart) and *Expense by Category* (stacked bars). We limited access to this report via Row-Level Security (RLS) so that only finance users could view sensitive data. This dashboard helps the CFO monitor financial health and identify cost overruns.

Staff Dashboard

The Staff dashboard covers **human resources** metrics. Visuals include:
- *Staff Count by Department* (column chart)
- *Vacancy Rate* (card)
- *Shift Utilization* (heat map of staffing by day/time).

It also tracks trends in hiring and attrition. For example, a DAX measure for attrition:

```
AttritionRate = DIVIDE(
    COUNTROWS(FILTER(Staff, Staff[Status]="Left")),
    COUNTROWS(Staff)
)
```

This calculates the percentage of staff who have left. Managers use this dashboard to ensure adequate staffing levels and to forecast hiring needs.

Technical Implementation Deep-Dive

The solution architecture combined **Power BI Desktop**, an **Azure SQL** (on-prem gateway), and Power BI Service. Key implementation details included:

- **Data Modeling:** We built a dimensional model (star schema) in Power BI. Fact tables (Admissions, Visits, Transactions) link to dimensions (Patients, Doctors, Date, Departments). The Date dimension supports time calculations (YTD, MTD). Relationships were enforced (one-to-many) and many-to-many scenarios were avoided by normalization. We followed best practices: measures are placed in a separate table (for clarity) and descriptive names used.

- **Power Query (M) Optimizations:** To improve performance, we enabled *query folding* where possible by letting transformations push down to SQL. Large tables used incremental refresh (monthly partition) in Power BI Service to avoid full loads. Example M code for a lookup merge:

  ```
  Source = Sql.Database("HospitalDB","Admissions"),
  AdmissionData = Source{[Schema="dbo",Item="Admissions"]}[Data],
  Join = Table.NestedJoin(AdmissionData, {"DoctorID"}, DoctorsTable,
  {"DoctorID"}, "DocInfo", JoinKind.LeftOuter),
  Expanded = Table.ExpandTableColumn(Join, "DocInfo", {"CommissionRate"},
  {"CommissionRate"})
  ```

  This merges the Doctors table to admission records for commission calculations.

> **Milestone:** We implemented Row-Level Security (RLS) early in development to ensure compliance. For example, a doctor role sees only their patient data, while executives see all departments [7].

- **DAX Measures:** Hundreds of measures were developed using DAX. We emphasized reusability and performance by using variables (`VAR`) and reducing filter context complexity. For instance, instead of repeating filter logic, we stored intermediate values in VAR and reused them. We also documented measures (in tooltips or comments) to clarify logic. For time intelligence, standard functions (e.g. `DATESYTD`, `PARALLELPERIOD`) were used with a Marked Date table.

- **Visual Design:** The report theme matched Capgemini's color palette. Icons and consistent fonts (Calibri) were used for readability. We limited the number of visuals per page to avoid clutter. Tooltips and drill-throughs provided additional details without crowding the main canvas.

Testing & QA

Quality assurance was integral. We performed:
- **Data Validation:** Every measure was verified by cross-checking against source queries. For example, total patient counts were validated by SQL count queries. DAX Studio was used to test performance and correctness of complex measures.
- **User Testing:** We followed a "real shift" testing approach [8]. Stakeholders (doctors, nurses, admins) were asked to use the dashboards in simulated scenarios: making decisions based on the visuals, finding flaws or missing metrics. They identified issues like filters not resetting or certain KPIs missing for sub-departments. We iteratively fixed these.
- **Performance Testing:** Large datasets (years of transactions) initially slowed report load. We optimized by implementing aggregations and reducing refresh schedules for slow-changing data (e.g. historical staff records were updated weekly instead of daily). The Power BI Performance Analyzer helped pinpoint slow visuals; we simplified their design.
- **Regression Testing:** After each sprint, we re-ran test cases to ensure new features didn't break existing functionality. We also tested the mobile layout to ensure reports rendered well on phones/tablets.

> **Tip:** Always test dashboards with real end-users. In our case, if a doctor couldn't glean necessary info "in 30 seconds during peak hours," we simplified the view [9]. This user-centric testing ensured the dashboards were intuitive and actionable.

Deployment & Training

- **Deployment:** Completed dashboards were published to Power BI Service workspaces. We created an App for easy access by users. Data sources were scheduled for refresh via an On-Premises Data Gateway (hourly for operations data, nightly for financials). Row-Level Security roles were applied on the dataset. We documented the data flow and refresh settings for maintainability.
- **Training:** We conducted live training sessions and created user guides covering dashboard navigation, filter usage, and drill-down features. For instance, users learned how to toggle between overview and detail filters. We also provided a short video demo (accessible on the intranet).
- **Support:** A feedback channel (Teams group) was established for ongoing support. After deployment, we held follow-up Q&A to ensure users were comfortable and to gather enhancement requests.

Challenges & Solutions

Several challenges arose, with corresponding solutions:
- **Inconsistent Data:** Department names and codes varied across sources. We resolved this by creating master lookup tables in Power Query to standardize values. This ensured visuals like *Admissions by Department* were accurate.
- **Performance Issues:** The initial query for billing data was very slow. We created indexed views in SQL and used query folding so that Power BI pulled pre-aggregated data. Incremental refresh eliminated long full-refresh times.
- **Changing Requirements:** Mid-project, the client requested new drill-downs (e.g., see commission by doctor name). We managed this with Agile iterations, adding new measures and bookmarks without derailing the timeline.
- **Data Security:** Ensuring only authorized users could see patient or financial data was critical. We implemented RLS based on Active Directory roles, as well as separate workspaces for finance vs clinical dashboards.
- **Avoiding Common Pitfalls:** We were conscious of known BI pitfalls. For example, we avoided dashboard bloat by limiting visuals per page, enforced consistent units/formats, maintained refresh schedules, and thoroughly tested with users [10]. This proactive approach averted issues such as missing updates or overwhelmed users.

Key Technical Skills

- **Power BI & DAX:** Designed complex reports with advanced measures (`CALCULATE`, `FILTER`, time-intel). Example:

```
CurrentPatients = CALCULATE(
    DISTINCTCOUNT(Patients[PatientID]),
    Patients[Status] = "Admitted"
)
```

- **Power Query (M):** Extracted and transformed data (joins, pivots, type casts). Example:

```
CleanData = Table.ReplaceErrorValues(RawData, {{"Charge", 0}})
```

- **Data Modeling:** Created star-schema models; built relationships between facts (Admissions, Billing) and dimensions (Patients, Doctors, Date).
- **SQL & ETL:** Wrote SQL queries and stored procedures to retrieve data; used Azure Data Factory for initial bulk loads.
- **Performance Tuning:** Utilized DAX Studio and query diagnostics; implemented aggregations and incremental refresh.
- **Security:** Configured Row-Level Security and workspace permissions in Power BI Service.
- **Agile Collaboration:** Used Jira for sprint tracking; coordinated with business analysts and QA.
- **Domain Tools:** Basic familiarity with healthcare systems (HL7, EHR terminology) and regulatory considerations (HIPAA).

Business Domain Knowledge

Through this project, I developed strong healthcare analytics expertise. Relevant domain insights included:
- **Healthcare KPIs:** Understanding metrics like ER wait times, bed occupancy, patient throughput, readmission rates, and length of stay. These are common in healthcare dashboards [11].
- **Clinical vs. Operational Needs:** Recognized the difference between clinicians (focus on patient care stats) and operations (focus on throughput/efficiency) [12].
- **Financial Processes:** Gained familiarity with hospital billing cycles, insurance claims, and cost centers. The Finance dashboard tracked claims submissions and denials to identify revenue leakages.
- **Staffing and Scheduling:** Learned how nurse/doctor rosters and shift coverage impact service levels (e.g. more admissions during day shifts). The Staff dashboard helped visualize these patterns.
- **Healthcare IT Systems:** Worked with common systems (EMR extracts, lab information systems) and ensured data integration without compromising privacy (using standardized anonymization where needed).

Project Outcomes & Impact

The completed analytics solution delivered significant value:
- **Efficiency Gains:** Automated reporting replaced dozens of static Excel reports. Executives now access live dashboards instead of waiting days for monthly reports. This cut decision-making time substantially.
- **Data-Driven Decisions:** Hospital leadership used our dashboards in daily meetings. For example, seeing high ER wait times directly led to reallocating staff in real time.
- **Accuracy & Transparency:** Centralizing data in Power BI improved accuracy (one source of truth) and transparency across departments. Discrepancies in prior manual reports were eliminated.
- **Adoption:** All 5 stakeholders (COO, CFO, head doctors) actively use the dashboards. Post-implementation surveys showed high satisfaction with the insights provided.
- **Reusable Framework:** The modular design (shared Date dim, measure tables) means new dashboards (e.g. Patient Satisfaction) can be built quickly. This established a **repeatable BI framework** for future projects.

Interview Prep Guide

This section highlights how to discuss this project in interviews:

- **Highlight End-to-End Role:** Emphasize that you managed the project from initial requirements through deployment. Be prepared to describe how you gathered requirements (stakeholder meetings, use-case definition), built data models (star schema), and designed visuals for each dashboard.
- **Discuss KPIs and Measures:** Interviewers may ask for specific examples. Mention key measures (e.g., calculating YTD revenue, readmission rate) and how you verified them. Be ready to explain a DAX formula or why you chose a particular visual.
- **Focus on Impact:** Explain how your work improved the hospital. For instance, “By integrating disparate data, we enabled real-time monitoring of ER capacity, which led to a 15% reduction in bottlenecks.” Concrete outcomes resonate well.
- **Tools & Techniques:** You should be able to talk about both Power Query (M) and DAX. For example, describe an M snippet used for data cleansing (such as merging or type conversion) and a DAX formula for a dynamic measure.

- **Challenges & Solutions:** Prepare examples of problems you overcame (data quality issues, performance). For each, describe the solution (e.g., implemented incremental refresh, created lookup tables for inconsistent codes). This shows problem-solving.
- **Domain Knowledge:** Mention any healthcare concepts you learned. For example, you might explain what bed occupancy means and why it matters, or how patient readmission within 30 days is a standard quality metric.
- **Soft Skills:** Highlight collaboration (working with doctors to identify metrics) and communication (presenting findings). Mention that you trained end-users and provided documentation.
- **Tableau vs Power BI:** If asked, note that Power BI was chosen for its Microsoft integration and cost-effectiveness [13]. You could say: "Power BI fit our MS ecosystem and licensing, whereas Tableau would have been more complex for our team."

  Tip: When discussing code or measures, speak clearly in business terms. Don't just recite formulas; explain *why* they matter (e.g., "This DAX measure calculates the average ER wait time, which was a critical metric for operations").

Tableau vs Power BI Decision

Early in the project, we evaluated both Power BI and Tableau. Key considerations: Power BI integrates seamlessly with Microsoft tools (Excel, Azure) and offers lower licensing costs [13]. Our team's background in SQL Server and Excel made Power BI familiar and easy to adopt. Tableau's strengths are its highly polished visualizations, but given our need for tight integration (and our budget constraints), Power BI was the optimal choice. In particular, we needed real-time connectivity to our SQL databases and a platform that users could quickly learn; Power BI met both requirements with its user-friendly interface and robust data source support.

Monthly Progress Summary

Month & Year	Key Activities and Milestones
Sept 2021	Project kickoff, stakeholder interviews, and requirement gathering.
Oct 2021	Environment setup: Installed Power BI tools, defined data sources, and began data analysis.
Nov 2021	Data profiling and initial ETL scripts; drafted conceptual data model and KPIs.
Dec 2021	Finalized schema (star model), developed sample data transformations, and reviewed wireframes with stakeholders.
Jan 2022	Developed **Overview** & **Patient** dashboards (first prototypes); implemented core DAX measures.
Feb 2022	Built **Doctor/Commission** and **Staff** dashboards; created commission and staffing calculations.
Mar 2022	Created **Operations** dashboard (ER, beds, labs); set up data refresh schedules (hourly/daily).

Month & Year	Key Activities and Milestones
Apr 2022	Developed **Finance** dashboard; implemented YTD revenue/expense measures and budget visuals.
May 2022	Internal QA: Debugged calculations, optimized slow visuals, and improved query performance.
Jun 2022	Published beta version to Power BI Service; held user acceptance (UAT) sessions; collected feedback.
Jul 2022	Iterated on feedback; implemented Row-Level Security (roles for clinicians, finance, executives); refined visual design.
Aug 2022	Conducted training workshops and webinars; distributed user manuals and cheat-sheets.
Sep 2022	Monitored usage metrics; performance tuning (adjusted refresh intervals, aggregated data).
Oct 2022	Added requested enhancements (e.g., additional drill-downs, report themes); finalized documentation.
Nov 2022	Project handover preparation: knowledge transfer to maintenance team, handed over SQL schemas.
Dec 2022	Official project closeout; final sign-off from stakeholders; planned next-phase enhancements.
Jan 2023 – Jun 2023	(Extended support) Resolved user issues, performed updates, and started planning new analytics features.

Key Takeaways & Next Steps

- **Technical Learning:** This project deepened my DAX and data modeling skills. I learned advanced patterns (e.g. variables in complex measures, incremental refresh setup). I also picked up healthcare domain knowledge (e.g. interpreting clinical KPIs) which helps in designing relevant reports.
- **Process Insight:** The importance of early stakeholder alignment was reinforced. Clear use cases (like targeting ER delays) guided development and avoided scope creep. Iterative feedback (Agile sprints) ensured the product met user needs.
- **Result:** We established a robust BI framework. As next steps, we plan to enhance predictive analytics (e.g., forecasting patient admissions with machine learning) and integrate more data (like patient satisfaction surveys) into the dashboards. The framework can be extended to new modules (e.g. Pharmacy or Radiology).
- **Career Growth:** This experience prepared me to discuss BI projects in interviews by showing end-to-end ownership. I can now confidently explain technical choices (why CALCULATE() was used, how RLS works) and how BI delivers business value. I aim to continue learning (e.g., Power BI certification) and apply these lessons to future healthcare or analytics projects.

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] Build a Power BI Healthcare Dashboard: Step-by-Step Guide
https://shivlab.com/blog/build-power-bi-healthcare-dashboard/

[13] Power BI Vs Tableau: Which is the Better BI Tool In 2026?
https://www.thoughtspot.com/data-trends/business-intelligence/power-bi-vs-tableau